

MicroPythonOS

FOSDEM – Sunday 1-2-2026

whoami



- Thomas Farstrike
- Creator of MPOS
- Background:
 - Kernel development
 - Embedded Linux
 - C/C++

Origins



- One Year Ago
- Building new device
- Need for modern ESP32 Touch OS
- ...with AppStore
- Didn't exist!

Create it in C/C++ ?



- Known territory
- Most ESP32 libraries
- Lightweight
- Logical choice
- Why not?

Create it in C/C++ ?



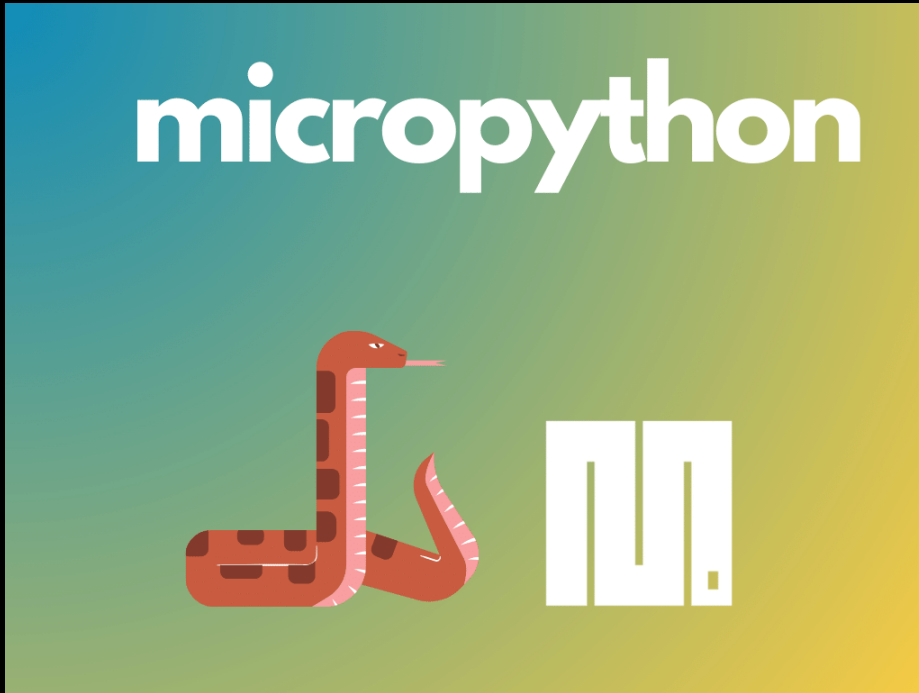
- Limited MMU
- Custom compilation
- Challenge to maintain
- Unforgiving
- Difficult to create OS
- Not easy for app developers

Create it in MicroPython!



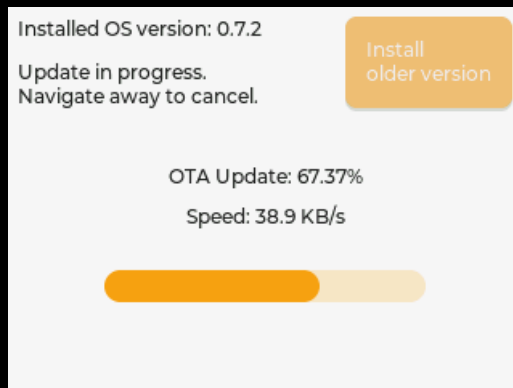
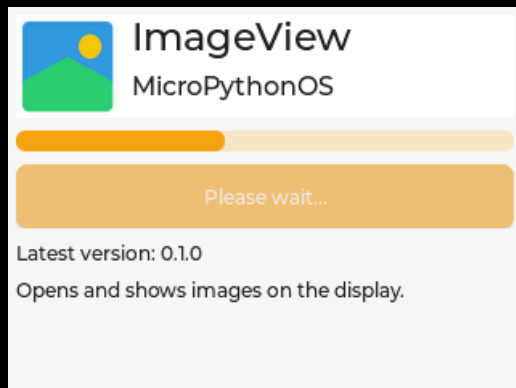
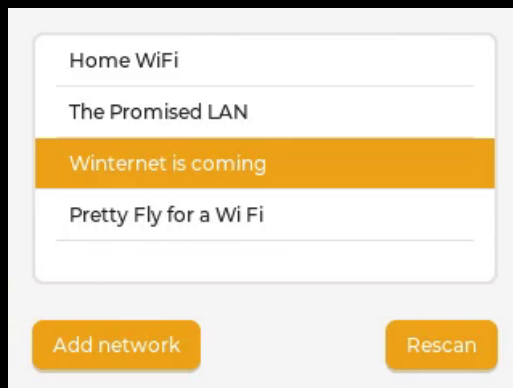
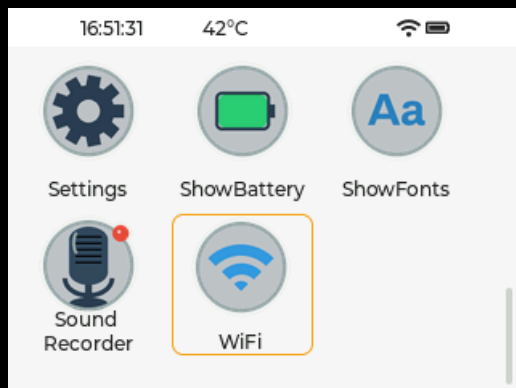
- Dynamic loading
- Scripts (.py)
- Binaries (.mpy)
- Modern language
- Easy for app developers
- Drivers in MicroPython

MicroPython



- Born 2014, ESP 2017
- Surprisingly fast
- Many libraries
- Also desktop (native)
- Quick Coding Cycle
- 99% like Python

Proving the concept



- Launcher
- WiFi + Auto-Connect
- AppStore
- OSUpdate
- Settings
- About

High-level architecture



- Apps
- Frameworks:
MicroPython and C
- High-level drivers:
MicroPython and C
- MicroPython Runtime
- Low-level drivers:
FreeRTOS, Darwin, Linux

LVGL makes it beautiful



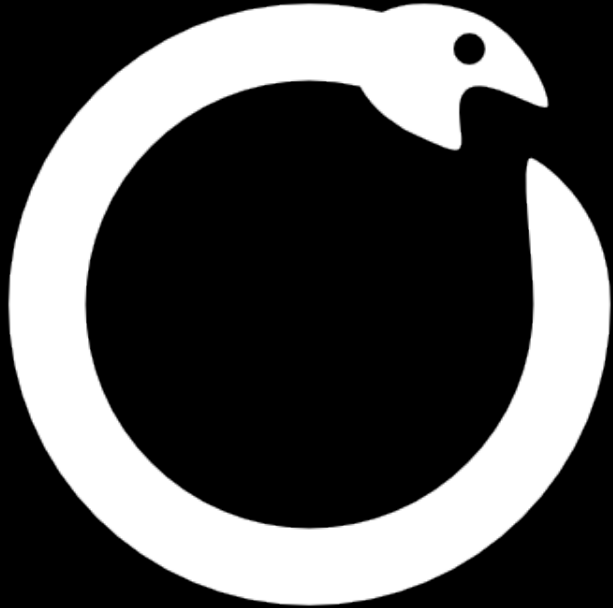
- Light and Versatile
 - 30+ Widgets
 - Animations
 - Image Decoders
 - UI Editor
 - MicroPython bindings
- Credit: lvgl_micropython

Lessons from Android



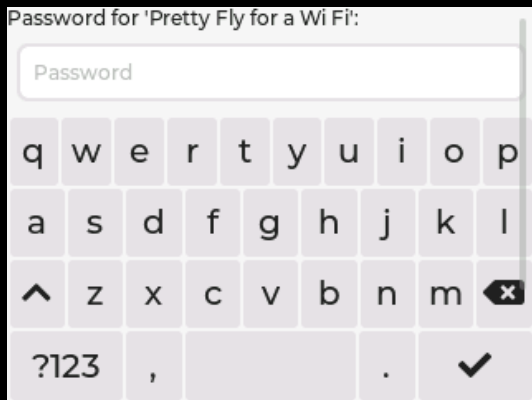
- Create once, run on all (compatible) devices
- Easy app development
- Frameworks
- User Interface

Why MicroPythonOS



- Very Lightweight
- Boots in 3 seconds
- Python language
- 100% open development
- Frequent releases
- Runs native on desktop
- Up-and-coming

Innovations



- MposKeyboard +50% buttons
- Swipe gestures for navigation
- Optimize QR decoding (960x960)
- v4l2 camera driver for MP
- Joystick for focus direction
- python-nostr ported to micropython-nostr
- Extend secp256k1 MP binding for ECDH

Apps

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Familiar app life cycle

```
hello.py* 1 from mpos import Activity
           2
           3 class Hello(Activity):
           4
           5     def onCreate(self):
           6         screen = lv.obj()
           7         label = lv.label(screen)
           8         label.set_text('Hello World!')
           9         label.center()
          10         self.setContentView(screen)
```

- Create Screen
- Add Label
- Set Label Text
- Center Label
- Activate Screen

Familiar MANIFEST.JSON

```
1 {  
2 "name": "HelloWorld",  
3 "publisher": "MicroPythonOS",  
4 "short_description": "Minimal app",  
5 "long_description": "Demonstrates the simplest app.",  
6 "full_name": "com.micropythonos.helloworld",  
7 "version": "0.0.2",  
8 "category": "development",  
9 "activities": [  
10     {  
11         "entrypoint": "assets/hello.py",  
12         "classname": "Hello",  
13         "intent_filters": [  
14             {  
15                 "action": "main",  
16                 "category": "launcher"  
17             }  
18         ]  
19     }  
20 ]  
21 }
```

- Name
- Publisher
- Descriptions
- Version
- Entrypoints

Familiar file structure

```
apps/com.micropythonos.helloworld/  
├── assets  
│   └── hello.py  
├── META-INF  
│   └── MANIFEST.JSON  
└── res  
    ├── mipmap-mdpi  
    │   └── icon_64x64.png
```

- Code (hello.py)
- MANIFEST.JSON
- Icon(s)
- Zipped to .mpk
- See docs

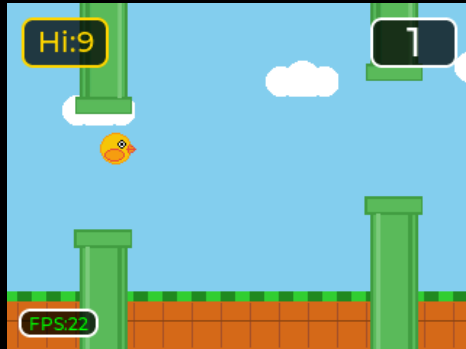
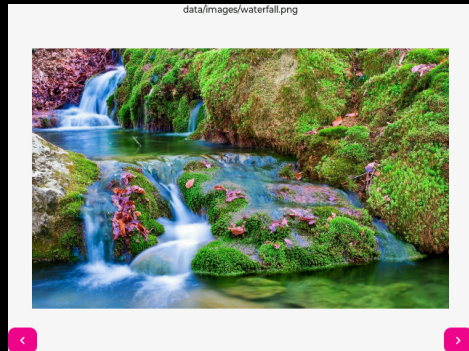
AppStore backends

The screenshot shows the BadgeHub web interface for submitting a new app. The top navigation bar includes links for 'Browse Projects', 'Create Project', 'My Projects', and 'API Docs', along with the user's name 'Thomas Farstrike'. The main form is divided into several sections:

- Basic Information:** Fields for 'App Name' (containing 'ImageView'), 'Author' (containing 'MicroPythonOS'), 'Git URL' (containing a GitHub link), 'Version' (containing '0.0.6'), and 'Description' (containing 'Image viewer that opens and shows images on the display. Allows deletion as well.'). There is also a 'Hidden' checkbox.
- Categorization:** Fields for 'Badge' (containing 'troopers23'), 'Category' (containing 'Graphics'), and 'License Type' (containing 'MIT').
- Files:** A section for uploading files, currently showing 'No files selected'.
- Project Files:** A list of files to be included in the app package, including 'com.micropythonos.imageview.0.0.6.apk', 'com.micropythonos.imageview.0.0.6.64x64.png', 'icon-64x64.png', and 'metadata.json'.

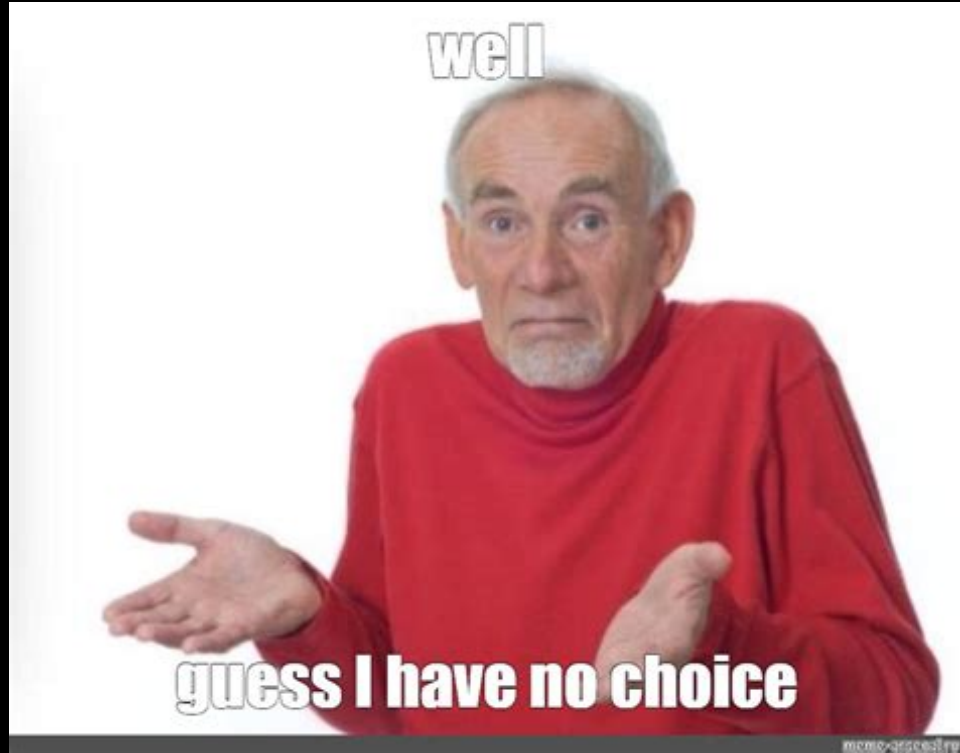
- apps.MicroPythonOS.com
 - Original
 - Curated List
 - To be listed: reach out
- Badge.WHY2025.org
to BadgeHub.eu
- More are welcome:
 - Centralized
 - Nostr (NIP-51, kind 32267)

Apps in the AppStore



- LightningPiggy
- Camera (with QR)
- Image Viewer
- Games
- Music Player
- Sound Recorder
- + 8 more

Why build a MicroPythonOS app



- Fun
- Easy (or Hard)
- Educational
- Many OSS examples
- Supportive community
- $1/1 > 1/100$
- Growing niche

Good documentation

MicroPythonOS Documentation

MicroPythonOS Documentation

Home

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Getting Started

Overview

Installation

Supported Hardware

Apps

Overview

Built-in Apps

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Creating Apps

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AppManager

AudioManager

BatteryManager

BuildInfo

ConnectivityManager

DeviceInfo

DisplayMetrics

DownloadManager

InputManager

LightsManager

Preferences

SensorManager

SettingActivity

SettingsActivity

TaskManager

TimeZone

WidgetAnimator

WifiService

Architecture

MicroPythonOS Documentation

Welcome to the official documentation for **MicroPythonOS**, a lightweight and versatile operating system built entirely in MicroPython. Designed for microcontrollers like the ESP32 and desktop systems, MicroPythonOS offers a modern, Android-inspired interface with an app ecosystem, App Store, and Over-The-Air (OTA) updates.

To see MicroPythonOS in action, check out the screenshots and screen videos on the [main website](#).

About MicroPythonOS

MicroPythonOS is tailored for developers and innovators building IoT devices, smart wearables, educational tools, and more. Its key features include:

- **Native MicroPython:** Write apps and drivers in MicroPython for rapid development.
- **Cross-Platform:** Runs on ESP32 microcontrollers, Linux desktops, and potentially MacOS and Raspberry Pi.
- **Modern UI:** Android-like touch screen interface with gestures.
- **App Ecosystem:** Built-in apps and an App Store for easy expansion.
- **OTA Updates:** Seamless system and app updates.

This documentation provides everything you need to get started, understand the architecture, build the system, and develop apps.

Who This Is For

- **Developers:** Learn how to install, build, and extend MicroPythonOS with new apps and features.
- **Managers:** Explore the system's capabilities, supported hardware, and ready-to-use features for prototyping and production.

Getting Started

New to MicroPythonOS? Start [here](#).

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About MicroPythonOS

Who This Is For

Getting Started

Contributing

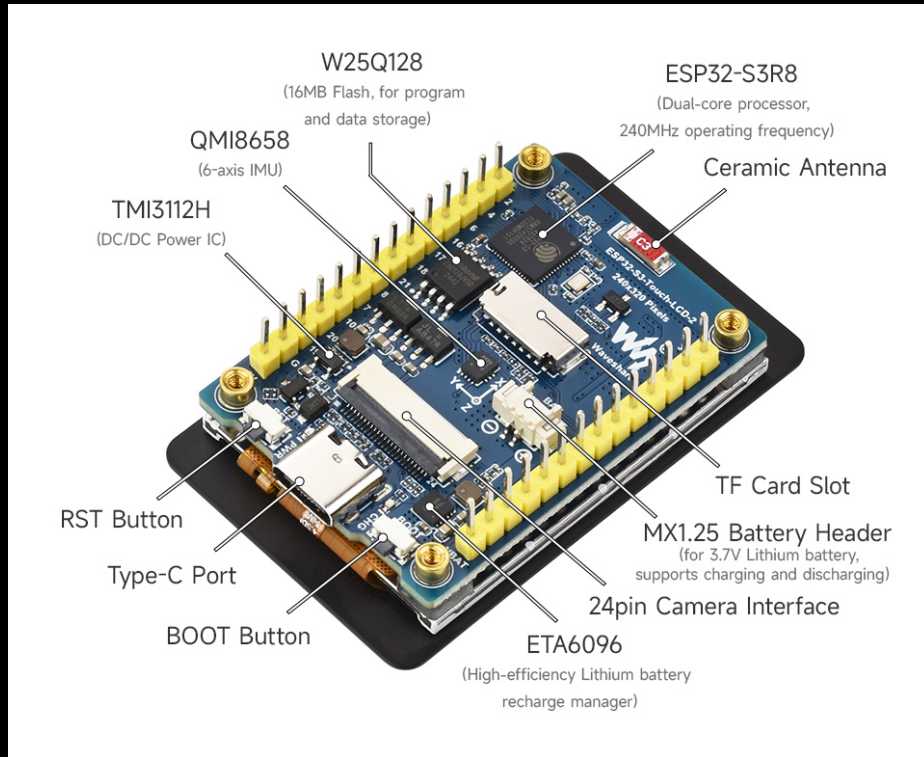
- All frameworks documented
- Updated regularly

<https://docs.MicroPythonOS.com>

Supported hardware

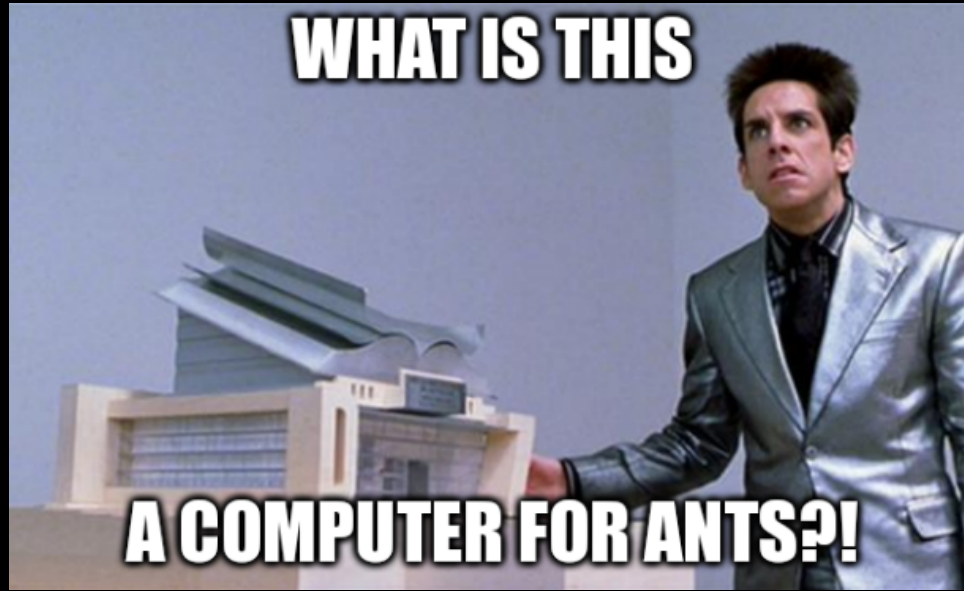
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First board: Waveshare S3 Touch 2



- 320x240
- 2 inch screen
- Capacitive Touch
- Camera 5MP
- Accelerometer (IMU)
- MicroSD Card
- Battery support

Hardware Specs



- 300KB SDRAM
- 8MB PSRAM
- 16MB ROM

Hardware Specs



- 300KB SDRAM
- 8MB PSRAM
- 16MB ROM

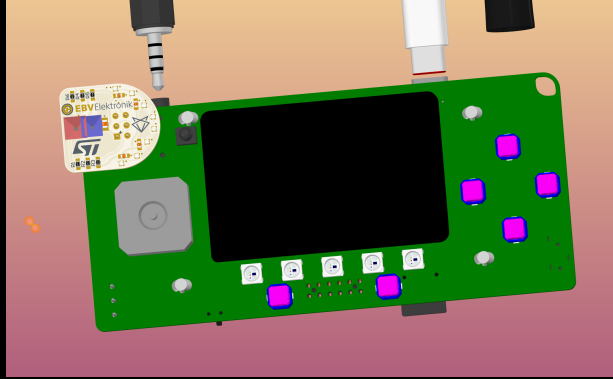
=~ IBM PS/2 (1981)

Second board: Fri3d Camp 2024



- 296x240
- 2 inch screen
- IMU
- Buzzer
- Multicolor LEDs
- Expansions

Third board: Fri3d Camp 2026



- 320x240 2 inch display
- Capacitive Touch!
- Audio Jack
- MicroPythonOS
- Retro-Go
 - Doom
 - Gameboy
 - NES

Next boards



- Manufacturers are interested – adds value
- 1st: samples received
- 2nd: samples shipped
- New form factors
- LoRa
- Porting for free
 - Hurry if you want it

Live Demo



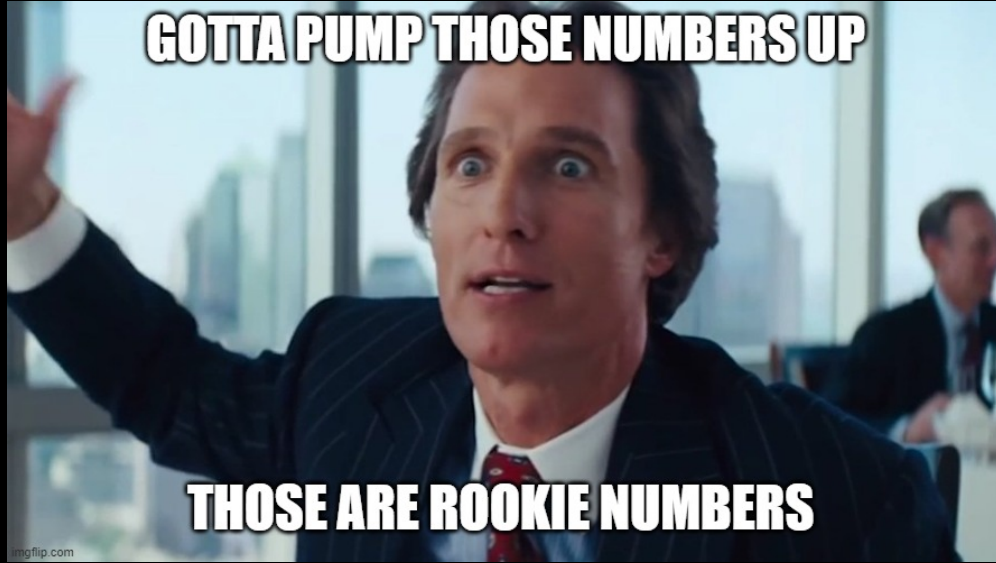
- Demo's are risky
- Demo's are scary
- Will it work?
- Where is it?

Live Demo



- You're looking at it!
- You've been it...
the whole time!

MicroPythonOS in Numbers



- Core Contributors: 4
- Watching: 6
- Apps: 15
- Forks: 20
- Community Chat: 53
- GitHub stars: 319
- Code commits: 2000

Roadmap



- Factory installs
- More frameworks
- API 1.0
- More drivers
- More boards
- Community driven

Next Steps: free stuff



- Run it on desktop
- Prebuilt binaries:
 - Linux
 - MacOS
- Build an app

Next Steps: buy stuff



- Get a supported device
- Flash it
- Install your app
- Publish in AppStore
 - Multiple backends

Easy Installation

Install MicroPythonOS

Click the button below to flash [MicroPythonOS](#) firmware onto your ESP device using WebSerial. Ensure your device is connected via USB and you're using a supported browser (e.g., Chrome or Edge).

Don't have a device yet? Buy a supported one, like the [Waveshare ESP32-S3-Touch-LCD-2](#). Or check out [the docs](#) to see how you can run it on your desktop PC.

Select Version

- Unified for all ESP32-S3 devices: Waveshare and Fri3d Camp 2024
Badge: version 0.7.1

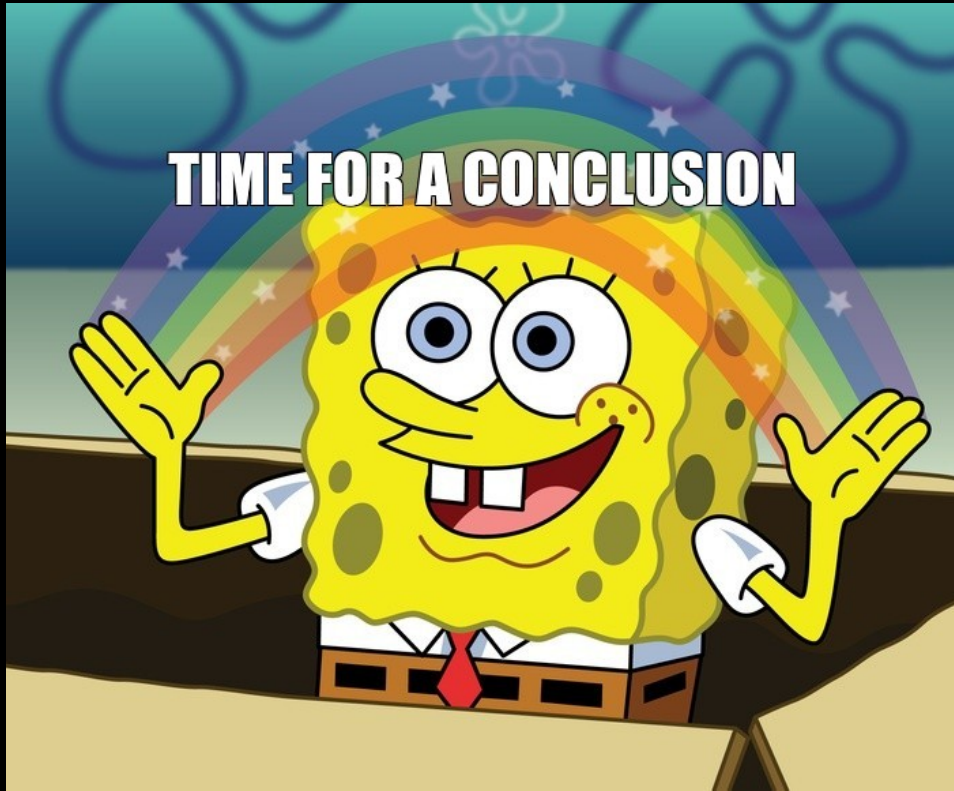
Show old releases ▼

Connect and Install

- WebSerial installer
- One firmware to rule them all

<https://install.MicroPythonOS.com>

Conclusion



- MicroPythonOS is waiting for you
- Join the chat
<https://chat.MicroPythonOS.com>
- Clone the repo
- Questions?

MicroPythonOS

Thank you!
